

(M₂) source-scope audit: Glimm–Jaffe–Spencer 1974

verified statements, explicit derivation, and the residual gates

Update (9 August 2026)

Gate R2 is closed by page-image inspection: the printed norm is $\|Q(x^0)\| = K_1[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)!]\|w\|_2$ — K_1 to the first power, but local weight $K_2^{2N(\Delta)}$. Consequently the constant correction “ $K_2^{4p} \rightarrow K_2^{2p}$ ” in Remark 5.3(i) below is *withdrawn* (an OCR artifact); the cell-count correction $32 \rightarrow 9$ stands. Exact constants: `r2-check.tex`.

Abstract

This note executes step 2 and step 5 of the execution order in `weak-coupling-strategy.tex`: the source-scope audit of the Glimm–Jaffe–Spencer estimates [1] against the copy in `refs/`, and the explicit derivation of the uniform local second moment (M₂) from them.

Audit outcome. Better than hypothesised, on four points, each verified against the printed text. (i) GJS’s Euclidean cutoff class is *verbatim* “ h with compact support in $\mathbb{R}^2$, $0 \leq h \leq 1$ ” — arbitrary, not unions of lattice squares — so the slab $h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g$ requires no extension of scope. (ii) Their *Hamiltonian* cutoff class (their (1.6)) is “ $0 \leq g \leq 1$, g measurable and of compact support”, identical to ours. (iii) The observable class (1.1.5) consists of *sharp-time* Wick monomials with L^2 kernels, with equal times explicitly allowed. (iv) The moment bound is their Theorem 1.1.8 and the cluster bound their Theorem 1.1.7, the latter stated with constants “independent of h ”.

Derivation outcome. $V(h)^2$ is *literally* a monomial sum of the class (1.1.5) with kernel $h \otimes h$ — no Wick recombination is needed — and the norm computation gives

$$\sup_{g \in \mathcal{G}} \|V(h)\Omega_g\|^2 \leq 9 C_8 \mathfrak{K}(p) \lambda^2 \left(\sum_{n=0}^p |a_n| \right)^2,$$

conditional on exactly one residual gate: the h -uniform reading of the constant in Theorem 1.1.8 (Gate 4.1). The slab transfer and the closedness limit are proved here from standard Feynman–Kac–Nelson and the spectral theorem; no uniform gap is used anywhere, so there is no circularity with (G). Two constants in `weak-coupling-strategy.tex` are corrected: $K_2^{4p} \rightarrow K_2^{2p}$ and $32 \rightarrow 9$.

A finding beyond the audit brief. GJS’s own display (1.1.12) differentiates the cutoff measure along the *interpolated family* $h_1 + \alpha g$ and combines it with Theorem 1.1.7 cell by cell: their proof of (1.1.11) is a Euclidean LPPL estimate with rate $e^{-m_1 d/2}$, at weak coupling, published in 1974. The state-derivative architecture of `lppl.tex` has an exact Euclidean precedent in the very source being audited.

1 What had to be verified

Hypothesis CE of `weak-coupling-strategy.tex` attributes to [1] two estimates, uniform over compactly supported cutoffs $0 \leq h \leq 1$: a moment bound (CE1) and a cluster bound (CE2), for sharp-time polynomials. The audit questions, in decreasing order of danger:

- Q1** Is the cutoff class in [1] really all of $\{h : 0 \leq h \leq 1, \text{supp } h \text{ compact}\}$, or only characteristic functions of unions of unit lattice squares? (If the latter, the slab $h_{T,g}$ with smooth g is outside literal scope and a variable-coupling extension would have to be proved.)
- Q2** Does the observable class admit *sharp-time* monomials — in particular *equal* times — with kernels controlled in a norm that tolerates $w = h \otimes h$? (An L^1 -in-time norm would die under sharp-time restriction.)
- Q3** Are the constants in the moment and cluster bounds uniform in h ?
- Q4** Do the conventions (covariance, Wick ordering, coupling normalisation) match `lpp1.tex`, or is a conversion of Wick masses needed?

2 Source audit

All quotations are from the JSTOR scan of [1] in `refs/`; page numbers are the printed Annals pagination 585–632. OCR-degraded formulas were re-derived from context where quoted.

Source Fact 2.1 (cutoff classes; Q1 resolved). p. 589: “Let h be a function with compact support in $\mathbb{R}^2$, and let $0 \leq h \leq 1$. Define the action $V(h) = \lambda \int : \mathcal{P}(\Phi(x)) : h(x) dx \dots$. Let dq_h be the cutoff measure”

$$dq_h = \frac{e^{-V(h)} d\Phi}{\int e^{-V(h)} d\Phi}.$$

No lattice-square restriction is imposed; rectangular cutoffs enter only as the particular sequence along which $dq = \lim_{h \uparrow 1} dq_h$ is constructed. Moreover, p. 588, their (1.6): the spatially cutoff Hamiltonian is $H(g, \lambda, m_0) = H_0(m_0) + \lambda \int : \mathcal{P}(\phi_0(x)) : g(x) dx$ with “ $\mathcal{P}(\cdot)$ bounded from below and $0 \leq g(\cdot) \leq 1$ with g measurable and of compact support”, and “ $H(g, \lambda, m_0)$ is essentially self-adjoint and inf spectrum H is a simple eigenvalue”. Both cutoff classes coincide with those of `lpp1.tex`; the slab $h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g$ is in the Euclidean class for every $T > 0$, $g \in \mathcal{G}$.

Source Fact 2.2 (slab limit stated in-source). p. 590, their (1.11): with $h_j = g(x^1) \chi_j(t)$, χ_j the characteristic function of $[-j, j]$, “the Feynman–Kac formula yields $W(x_1, \dots, x_n) = \lim_{j \rightarrow \infty} S_{h_j}(x_1, \dots, x_n)$ ”, where W are the vacuum expectation values of the *spatially cutoff Hamiltonian theory* (their (1.7)) and S_h the moments of dq_h . This is precisely the $T \rightarrow \infty$ slab transfer, asserted at the level of moments in the source itself.

Source Fact 2.3 (observable class and norms; Q2 resolved). pp. 593–594, their (1.1.5)–(1.1.9): the class consists of monomials

$$Q(x^0) = \int d\vec{x} w(\vec{x}) \prod_{i=1}^r : \Phi(x_i)^{m_i} :, \quad Q = \int dx^0 Q(x^0),$$

with $x_i = (x_i^0, x_i^1)$, the *times* x_i^0 fixed and only the space variables integrated against w ; Q is localized if w is supported in a product of unit lattice squares $\Delta_{j_1} \times \dots \times \Delta_{j_r}$; the degrees obey $m_i \leq \kappa$ with (p. 593, footnote 9) “any choice $\kappa \geq \deg \mathcal{P}$ is sufficient”. With $N(\Delta)$ the local degree, the norms are

$$\|Q(x^0)\| = K_1^e \prod_{\Delta} K_2^{N(\Delta)} N(\Delta)! \|w(x^0, \cdot)\|_{L^2}, \quad \|Q\| = \sum_l \|Q_l\| \quad (1.1.9), \text{ sum over localized pieces,}$$

where the exponent e on K_1 is not legible in the scan (Gate 4.2). Crucially the kernel norm is L^2 in the space variables at fixed times. Equal times are explicitly admitted: p. 596, “we take all times $x_1^0 = \dots = x_r^0$ equal in $Q(x^0)$. To indicate this restriction, we write $Q(t)$ ”.

Source Fact 2.4 (the two estimates; Q3). p. 594, Theorem 1.1.7: “There are constants $0 < m_1$ and $O(1)$ **independent of** h , Q_1 and Q_2 such that $|\int Q_1 Q_2 dq_h - \int Q_1 dq_h \int Q_2 dq_h| \leq O(1)e^{-m_1 d} \|Q_1\| \|Q_2\|$ ” (d the widest separating strip). This is CE2, with h -uniformity in the statement.

p. 595, Theorem 1.1.8: “For $Q(x^0)$ of the form (1.1.5), $|\int Q(x^0) dq_h| \leq O(1)\|Q(x^0)\|$ **uniformly as** $h \rightarrow 1$ ”. This is CE1. The phrase “uniformly as $h \rightarrow 1$ ” is the one point where the printed statement is weaker than the class-uniform reading; see Gate 4.1 for why the paper’s internal logic nevertheless commits to the class-uniform version.

p. 596, Theorem 1.1.11: the n -particle decoupling estimate, with (1.1.16) asserted “uniformly in h ”.

Source Fact 2.5 (the engine). p. 615, Theorem 2.4.1: “Let s and n be given. Let m_0 , K_2 and K_1 be sufficiently large and let λ be sufficiently small. Then the increasing contour expansion (2.3.7) is norm decreasing”, with the constants chosen in the order m_0, K_2, K_1, λ — none of them a function of h . The expansion (their §2.0–2.2) differentiates only the interpolated covariances $C(s)$, never h ; the cutoff rides inside the vertices $-\lambda h(x) : \mathcal{P}'(\Phi(x)) :$ produced by $\delta/\delta\Phi$ acting on $e^{-V(h)}$.

Source Fact 2.6 (GJS’s own Euclidean LPPL). p. 595, (1.1.11)–(1.1.12): for cutoffs h_1, h_2 and Q supported in $\mathcal{O}$, with $g = h_2 - h_1$, $g_i = g\chi_{\Delta_i}$,

$$\langle Q, \rangle_{h_2} - \langle Q, \rangle_{h_1} = - \int_0^1 d\alpha \sum_{i \in \mathbb{Z}^2} \left[\langle QV(g_i), \rangle_{h_1 + \alpha g} - \langle Q, \rangle_{h_1 + \alpha g} \langle V(g_i), \rangle_{h_1 + \alpha g} \right],$$

and applying Theorem 1.1.7 to each term, $|\langle Q, \rangle_{h_2} - \langle Q, \rangle_{h_1}| \leq O(1)e^{-m_1 d/2} \|Q\|$, $d = \text{dist}(\mathcal{O}, \text{supp}(h_1 - h_2))$. Three observations. (i) The interpolated cutoffs $h_1 + \alpha g$ are not characteristic functions even when h_1, h_2 are; the paper therefore *uses* Theorems 1.1.7–1.1.8 on the full class of Source Fact 2.1. (ii) Structurally this is the state-derivative-plus-cell-expansion of `lppl.tex`, executed on the Euclidean side. (iii) It is the direct precedent for the claim that the connected correlator against a far-away interaction cell is the right object; the Hamiltonian-side novelty of `lppl.tex` is the sharp-time/operator-norm conclusion, not the mechanism.

Proposition 2.7 (conventions; Q4 resolved, no conversion needed). *The Wick ordering in $V(h)$ of Source Fact 2.1 is with respect to $d\Phi$, the Gaussian with covariance $G = (-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$ (their p. 589, (1.8)). The interaction of `lppl.tex` is Wick ordered with respect to the time-zero covariance $C = (2\mu)^{-1}$. These agree at sharp time: by `t3.tex`, Lemma 2.1, $G((0, x), (0, y)) = C(x - y)$ exactly, so every contraction constant entering the two Wick orderings of a sharp-time monomial is the same number, and*

$$: \Phi(0, x)^n :_{d\Phi} = : \phi_0(x)^n :_C$$

as L^2 limits of the identical mollified expressions. The coupling normalisation agrees (λ multiplies the full Wick polynomial in both), and the additive energy shift is absent from dq_h by construction (the measure is normalised by division). No analogue of the Wick-mass conversion (6.1) of `weak-coupling-strategy.tex` is required: the source Wick mass is m_0 .

3 Derivation of (M₂)

Standing assumptions: $\mathcal{P}(\xi) = \sum_{n=0}^p a_n \xi^n$ bounded below; λ/m_0^2 in the GJS weak-coupling region; h real with $\|h\|_\infty \leq 1$ and $\text{supp } h$ contained in an interval of length 2; $g \in \mathcal{G}$;

$$V(h) = \lambda \int h(x) : \mathcal{P}(\phi_0(x)) :_C dx, \quad K_g = H_0 + \lambda \int g : \mathcal{P}(\phi_0) :_C dx.$$

C_8 denotes the constant of Theorem 1.1.8 in its class-uniform reading (Gate 4.1); $\mathfrak{K}(p)$ the largest (1.1.8)-prefactor of a localized two-point monomial of per-point degree $\leq p$, so that, for every legible reading of the K_1 -exponent,

$$\mathfrak{K}(p) \leq K_1^{e_*} K_2^{2p} (2p)!, \quad e_* \leq \max(2, 2p).$$

Lemma 3.1 (class membership and norm). *On path space define the equal-time monomial sum at time 0*

$$Q_V := \lambda^2 \sum_{n,n'=0}^p a_n a_{n'} \iint h(x) h(y) : \Phi(0, x)^n :_{d\Phi} : \Phi(0, y)^{n'} :_{d\Phi} dx dy.$$

Then Q_V is a sum of monomials of the class (1.1.5) with $r = 2$, equal times, kernels $a_n a_{n'} h \otimes h \in L^2$, and per-point degrees $\leq p$, admissible with $\kappa = \deg \mathcal{P}$ by footnote 9. Its norm (1.1.9) obeys

$$\|Q_V\| \leq 9 \mathfrak{K}(p) \lambda^2 \left(\sum_{n=0}^p |a_n| \right)^2.$$

Proof. Membership is by inspection — the product of two individually Wick-ordered powers at distinct points is the form (1.1.5); no recombination into jointly ordered monomials occurs, so no contraction constants appear. For the norm: $\text{supp } h$ meets at most 3 unit lattice cells; write $h = \sum_{j \in J} h_j$, $h_j = h \chi_{[j, j+1]}$, $|J| \leq 3$, $\|h_j\|_{L^2} \leq 1$ since $|h| \leq 1$. Each localized piece $Q_{n,n'}^{(j,j')}$ has kernel $a_n a_{n'} h_j \otimes h_{j'}$ with $\|h_j \otimes h_{j'}\|_{L^2} = \|h_j\|_2 \|h_{j'}\|_2 \leq 1$, and local degrees: if $j = j'$ one square carries $N(\Delta) = n + n' \leq 2p$, contributing $K_2^{n+n'} (n + n')! \leq K_2^{2p} (2p)!$; if $j \neq j'$ two squares carry n and n' , contributing $K_2^n n! K_2^{n'} n'! \leq K_2^{2p} (p!)^2 \leq K_2^{2p} (2p)!$. Hence $\|Q_{n,n'}^{(j,j')}\| \leq |a_n| |a_{n'}| \mathfrak{K}(p)$, and (1.1.9) sums the ≤ 9 pairs (j, j') and the degrees. $\square$

Lemma 3.2 (slab Feynman–Kac–Nelson). *Let $T > 0$, $g \in \mathcal{G}$, $\psi_T := e^{-TK_g} \Omega_0$, $\hat{\psi}_T := \psi_T / \|\psi_T\|$. Then for every measurable $F : \mathcal{S}'(\mathbb{R}) \rightarrow [0, \infty]$,*

$$\int F(\Phi(0, \cdot)) dq_{h_T, g} = \frac{\langle \psi_T, F(\phi_0) \psi_T \rangle}{\|\psi_T\|^2};$$

in particular, with $F = V(h)^2$ (a nonnegative measurable function of the time-zero field, by Proposition 2.7),

$$\|V(h) \hat{\psi}_T\|^2 = \int Q_V dq_{h_T, g}.$$

Proof. For bounded F this is the standard Feynman–Kac–Nelson identity for the measure $Z^{-1} e^{-V(h_T, g)} d\Phi$: the Markov property of $d\Phi$ with respect to constant-time lines and $\int_{-T}^T \int g : \mathcal{P} : dx dt = \int_{-T}^T V(g)(t) dt$ give $\int F e^{-V(h_T, g)} d\Phi = \langle \Omega_0, e^{-TK_g} F e^{-TK_g} \Omega_0 \rangle$ (Nelson; [3, Thm. III.12]; [2]); any additive normalisation of K_g cancels in the ratio. Both sides are increasing in F , so monotone convergence extends the identity to $F \geq 0$ measurable, as an identity in $[0, \infty]$. The second display is the case $F = V(h)^2$: sharp-time Wick powers with respect to $d\Phi$ coincide with the C -Wick powers of ϕ_0 by Proposition 2.7, and then $F(\phi_0) = V(h)^2$ as a multiplication operator on Q -space. $\square$

Lemma 3.3 (limit; no uniform gap used). $\hat{\psi}_T \rightarrow \Omega_g$ in norm as $T \rightarrow \infty$. Consequently, if $\sup_T \|V(h)\hat{\psi}_T\| \leq B < \infty$, then $\Omega_g \in D(V(h))$ and $\|V(h)\Omega_g\| \leq B$.

Proof. By Source Fact 2.1 (or GJ II for smooth g), $E_g = \inf \text{spec } K_g$ is a simple isolated eigenvalue with eigenvector $\Omega_g > 0$ a.e. in Q -space; let $\gamma_g > 0$ be its (g -dependent, not uniform) distance to the rest of the spectrum. The spectral theorem gives

$$e^{-T(K_g - E_g)}\Omega_0 = c\Omega_g + r_T, \quad c = \langle \Omega_g, \Omega_0 \rangle = \int \Omega_g d\phi_0 > 0, \quad \|r_T\| \leq e^{-T\gamma_g},$$

whence $\hat{\psi}_T \rightarrow \Omega_g$. For the second claim: $V(h)$ is multiplication by a real function in $L^2(d\phi_0)$, hence self-adjoint on its maximal domain, hence closed; a closed graph is a norm-closed subspace and therefore weakly closed. If $\sup_T \|V(h)\hat{\psi}_T\| \leq B$, pick a subsequence with $V(h)\hat{\psi}_{T_k} \rightharpoonup v$; then $(\hat{\psi}_{T_k}, V(h)\hat{\psi}_{T_k}) \rightharpoonup (\Omega_g, v)$ lies in the graph, so $\Omega_g \in D(V(h))$, $V(h)\Omega_g = v$, and $\|v\| \leq \liminf \|V(h)\hat{\psi}_{T_k}\| \leq B$. $\square$

Theorem 3.4 ((M₂), conditional on Gate 4.1 only). In the GJS weak-coupling region, for every real h with $\|h\|_\infty \leq 1$ and support in an interval of length 2,

$$\sup_{g \in \mathcal{G}} \|V(h)\Omega_g\|^2 \leq 9C_8 \mathfrak{K}(p) \lambda^2 \left(\sum_{n=0}^p |a_n| \right)^2 =: B^2.$$

Proof. Fix g and T . The slab cutoff $h_{T,g}$ lies in the class of Source Fact 2.1, so Theorem 1.1.8 (class-uniform reading) and Lemma 3.1 give

$$0 \leq \int Q_V dq_{h_{T,g}} \leq C_8 \|Q_V\| \leq B^2,$$

the left inequality because $\int Q_V dq = \|V(h)\hat{\psi}_T\|^2 \geq 0$ by Lemma 3.2. Thus $\sup_T \|V(h)\hat{\psi}_T\| \leq B$, and Lemma 3.3 concludes. Uniformity in g holds because B contains no g -dependent quantity; translation covariance (m0, Lemma 3.2) makes the bound independent of the location of $\text{supp } h$ as well. $\square$

Remark 3.5 (free-field anchor). For $\lambda = 0$ the exact value is $\|V(h)\Omega_0\|^2 = \lambda^2 \sum_n a_n^2 n! \langle h, C^{on} h \rangle$ (cross degrees vanish), finite for every n since $C^{on} \in L_{\text{loc}}^1$; numerically, for $m_0 = 1$ and $h = \chi_{[0,1]}$: $\langle h, C^{on} h \rangle = 0.2687, 0.1008, 0.0509, 0.0326$ for $n = 1, \dots, 4$ (e.g. $\mathbb{E}[F_4[h]^2] = 4! \cdot 0.0326 = 0.78$). The bound of Theorem 3.4 sits far above these values, as it must; the point of the anchor is that the sharp-time second moments the theorem controls are finite non-vacuous quantities already at $\lambda = 0$.

4 Residual gates

Residual gate 4.1 (the h -uniform reading of Theorem 1.1.8). The printed statement says “uniformly as $h \rightarrow 1$ ”; the derivation above uses “uniformly over $0 \leq h \leq 1$, $\text{supp } h$ compact”. Three facts pin the stronger reading to the paper’s own logic, short of a line-by-line reconstruction of §§2–4: (i) Theorem 1.1.7 is stated class-uniformly (“independent of h ”), and its proof (p. 596–597) consists of applying Theorems 1.1.8 and 1.1.11 — an h -dependent C_8 would break the derivation; (ii) the proof of (1.1.11) applies Theorem 1.1.7 to the interpolated cutoffs $h_1 + \alpha g$, which sweep the full class even when h_1, h_2 are rectangles; (iii) in the expansion, h enters only multiplicatively inside vertices ($|\lambda h(x)| \leq \lambda$) and through stability factors $e^{-V(h)}$, and the constants of Theorem 2.4.1 are fixed in the order

m_0, K_2, K_1, λ with no reference to h . *What remains to be checked in a clean copy:* that the stability estimates used in §2.4/§3 (the analogues of $\|e^{-V_\Delta(h)}\|_{L^p(d\Phi)} \leq e^{O(1)}$ per unit square) are written with constants depending on h only through $0 \leq h \leq 1$ — the OCR of those pages is too degraded to certify each display. This is a bounded verification task, not an open problem.

Residual gate 4.2 (the K_1 exponent). The exponent e in the norm (1.1.8) prefactor K_1^e is illegible in the scan (readings $e \in \{1, r, \sum N(\Delta)\}$ are all consistent with the visible text). This affects only the numerical size of $\mathfrak{K}(p)$ through $e_* \leq \max(2, 2p)$, not the finiteness or the g -uniformity of Theorem 3.4. To be fixed by inspection of a clean copy.

Remark 4.3 (corrections to `weak-coupling-strategy.tex`). (i) The claimed constant $32\lambda^2 C_{\text{CE}} K_1 K_2^{4p} (2p)! (\sum |a_n|)^2$ should read as in Theorem 3.4: the cell count for a length-2 support is 9, not 32, and the local degrees sum to $n + n' \leq 2p$, giving K_2^{2p} , not K_2^{4p} . Both corrections go in the favourable direction. (ii) The citation “Theorems 1.1.7–1.1.8” for the pair (moments, cluster) should be (Thm. 1.1.8, Thm. 1.1.7) respectively; the numbering itself is confirmed correct. (iii) The Wick-mass conversion step (their (6.1)) can be deleted for this source: by Proposition 2.7 the source Wick mass is m_0 and the sharp-time orderings coincide exactly.

Remark 4.4 (what this does *not* settle). Nothing here bears on (G): the gap derivation is a separate transfer (CE2 on a dense cylinder core, then the spectral-measure lemma of `remaining.tex`), and remains open pending the same kind of audit for the time-separated case, plus the density lemma. The present note settles the (M₂) side conditional on Gate 4.1, which is shared by both transfers.

5 Status

- **Verified in source** (with page references): the cutoff classes (Euclidean and Hamiltonian) coincide verbatim with ours; sharp equal-time monomials with L^2 kernels are in the observable class; Thm 1.1.7 is class-uniform in h by statement; the slab limit is asserted in-source; the conventions match exactly, eliminating the Wick-mass conversion.
- **Proved here:** Lemmas 3.1–3.3 and Theorem 3.4 — (M₂) with the explicit constant $9 C_8 \mathfrak{K}(p) \lambda^2 (\sum |a_n|)^2$, uniform in g , using no uniform gap.
- **Remaining:** Gates 4.1 and 4.2 — both bounded verification tasks against a clean copy of [1], not open problems.
- **Bonus:** Source Fact 2.6 — GJS (1.1.11)–(1.1.12) is a 1974 Euclidean LPPL with rate $e^{-m_1 d/2}$, by exactly the derivative-plus-cell mechanism of `lppl.tex`.

References

- [1] J. Glimm, A. Jaffe and T. Spencer, *The Wightman axioms and particle structure in the $\mathcal{P}(\phi)_2$ quantum field model*, Ann. of Math. **100** (1974) 585–632.
- [2] F. Guerra, L. Rosen and B. Simon, *The $P(\phi)_2$ Euclidean quantum field theory as classical statistical mechanics*, Ann. of Math. **101** (1975) 111–189, 191–259.
- [3] B. Simon, *The $P(\Phi)_2$ Euclidean (Quantum) Field Theory*, Princeton University Press, 1974.